Modelling Global Life Expectancy from Health and Socioeconomic Factors Using Multiple Linear Regression

Course: STAC67 | Instructor: Dr. So-hee Kang

2025-12-02

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Abstract

Life expectancy is a key indicator of public health and socioeconomic development, shaped by factors such as disease burden, healthcare access, nutrition, and economic stability. This project aimed to build a statistically robust model to predict life expectancy across countries using WHO data from 2000–2015. After removing implausible values, high-missingness cases, and extreme outliers, we conducted exploratory data analysis and addressed multicollinearity using a systematic VIF-based elimination procedure. We developed several regression models, beginning with a main-effects model and progressing to an interaction model and finally a polynomial interaction model that incorporated nonlinear patterns and biologically meaningful interactions. Model comparison using R^2, R^2_{adj}, C_p, AIC, BIC, and PRESS identified the polynomial interaction model as the best-performing and most parsimonious. The model revealed strong effects of income distribution, deaths per 1,000 live births due to HIV/AIDS (ages 0–4), childhood health, and health expenditure, along with a nonlinear influence of adult mortality and key interactions involving HIV and childhood thinness. We

acknowledge limitations related to missingness and filtering decisions, and propose methodological extensions for future research.

Background and Significance

Life expectancy is a central indicator of population health and national development, reflecting the combined effects of healthcare access, economic conditions, vaccination coverage, nutrition, and disease burden. Countries with strong healthcare systems and equitable resource distribution tend to achieve longer lifespans, while those facing high infectious disease rates or inadequate infrastructure often experience shorter ones. Because of this, policymakers, researchers, and insurers rely on life expectancy to guide health interventions and risk assessments.

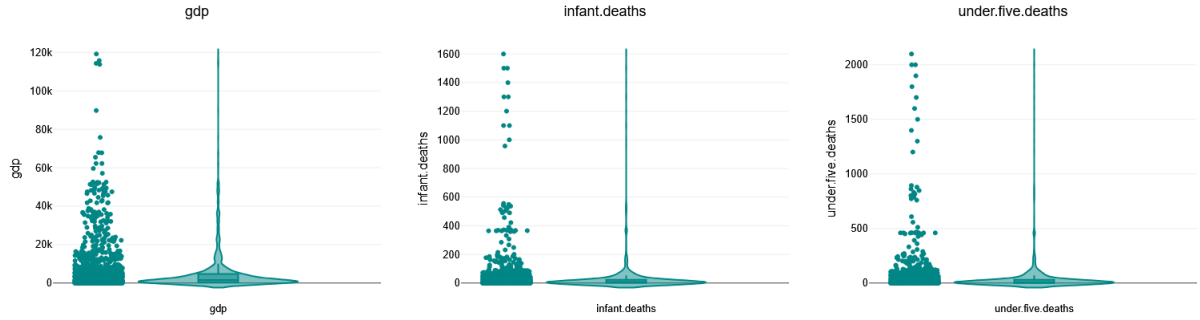
Modeling life expectancy, however, is challenging. The underlying predictors are highly interdependent, often nonlinear, and frequently contain skewness, multicollinearity, or missingness—issues especially visible in the WHO global dataset. This complexity requires thoughtful preprocessing and advanced modeling strategies beyond simple linear regression.

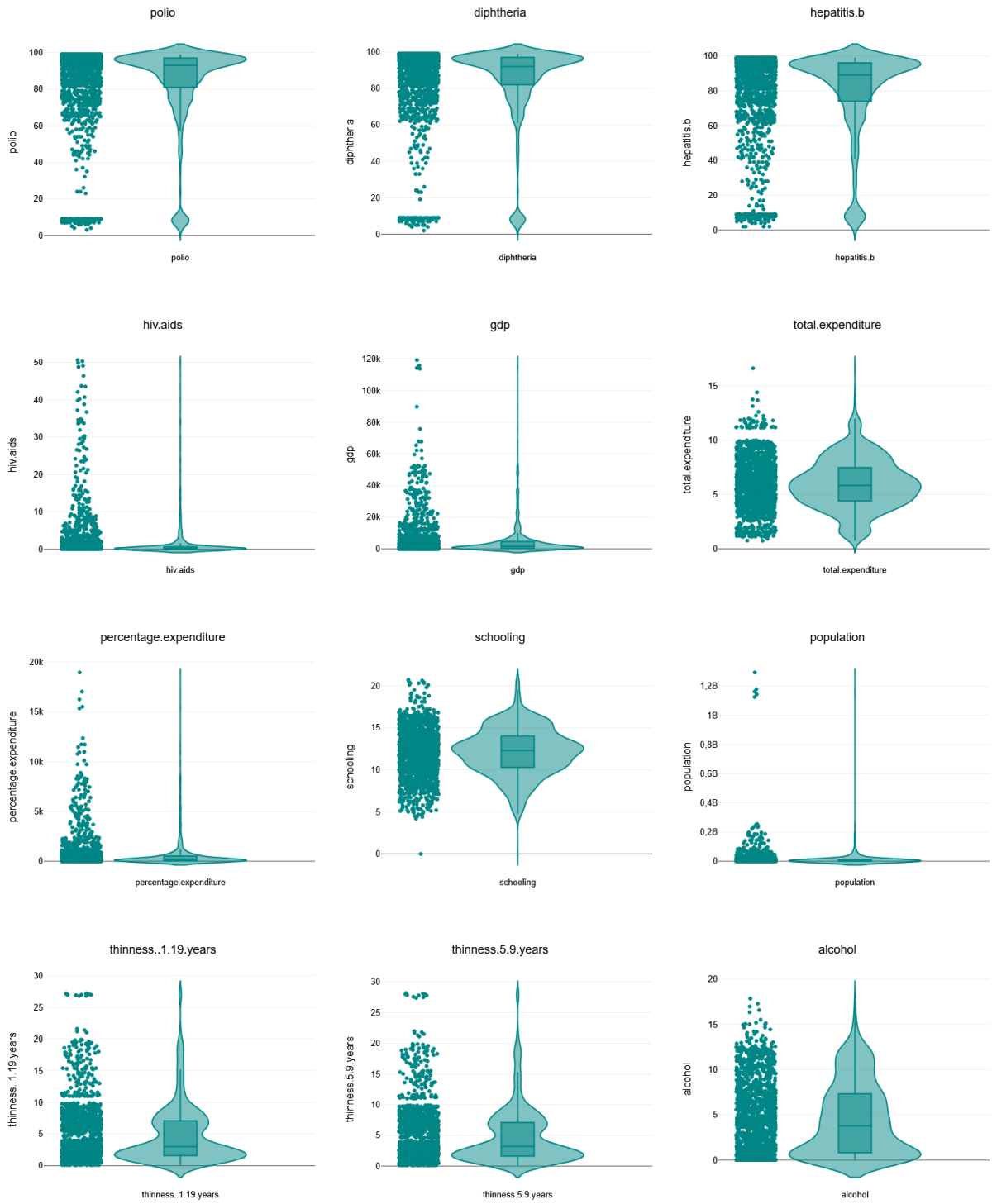
This project aims to address this challenge through the construction of increasingly sophisticated regression models that incorporate transformations, interactions, and nonlinear structure. Our goal is to answer the core question: **To what extent do socioeconomic, demographic, and health indicators predict global life expectancy, and which statistical modeling approach best captures these relationships?** The following sections describe our data preparation, exploratory analysis, and progression from main-effects to interaction and polynomial interaction models, culminating in the selection of the final model.

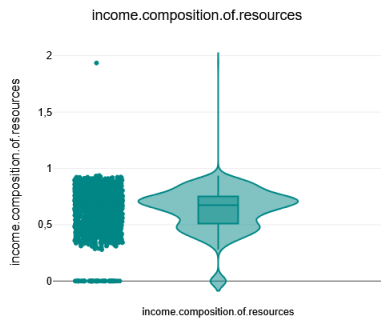
Exploratory Data Analysis

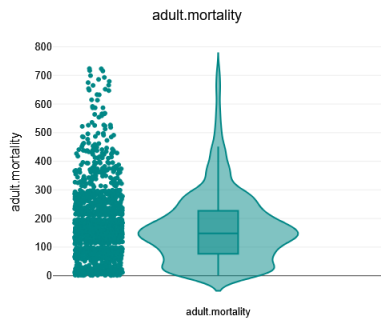
The dataset contains about 2,900 observations with life expectancy as the response and a wide set of health, economic, demographic, and healthcare-related predictors. Our exploratory analysis focused on examining distributions and identifying impossible or inconsistent values.

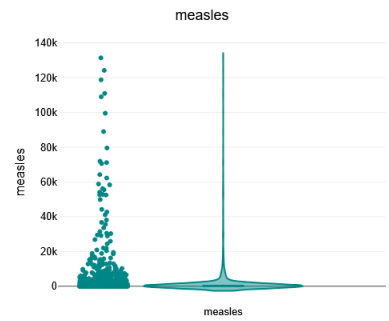
Several variables showed severe data quality issues. BMI contained biologically impossible values (e.g., <6.7 or >60), affecting over 20% of observations, so it was removed. Some rate variables (infant deaths, under-five deaths, measles) exceeded 1000 per 1000 population and were deleted as impossible values. Population also contained extreme errors (e.g., very small values for large countries), so observations below 100,000 were excluded. Missingness was substantial for variables like Hepatitis B, GDP, and Population (>15%), and since the missingness was not random, cases with missing values were removed to avoid biased imputations.




polio
diphtheria
hepatitis.b
hiv.aids
gdp
total.expenditure
percentage.expenditure
schooling
population
thinness..1.19.years
thinness.5.9.years
alcohol







We used violin plots and histograms to examine the distribution of quantitative variables. Several variables—such as under-five deaths, deaths per 1,000 live births due to HIV/AIDS (ages 0–4), population, and health-care expenditure—were strongly right-skewed, indicating a need for log-transformations. In contrast, variables such as the thinness indicators showed relatively symmetric distributions and did not require transformation.

To address multicollinearity, we used VIF rather than simple correlations, since VIF captures joint collinearity across predictors. Using a threshold of 4, we iteratively removed variables with the highest VIF-such as infant deaths, GDP, hepatitis B, polio, thinness 1–19, and schooling-while keeping those most strongly related to life expectancy. This produced a stable predictor set with all VIF values below the threshold.

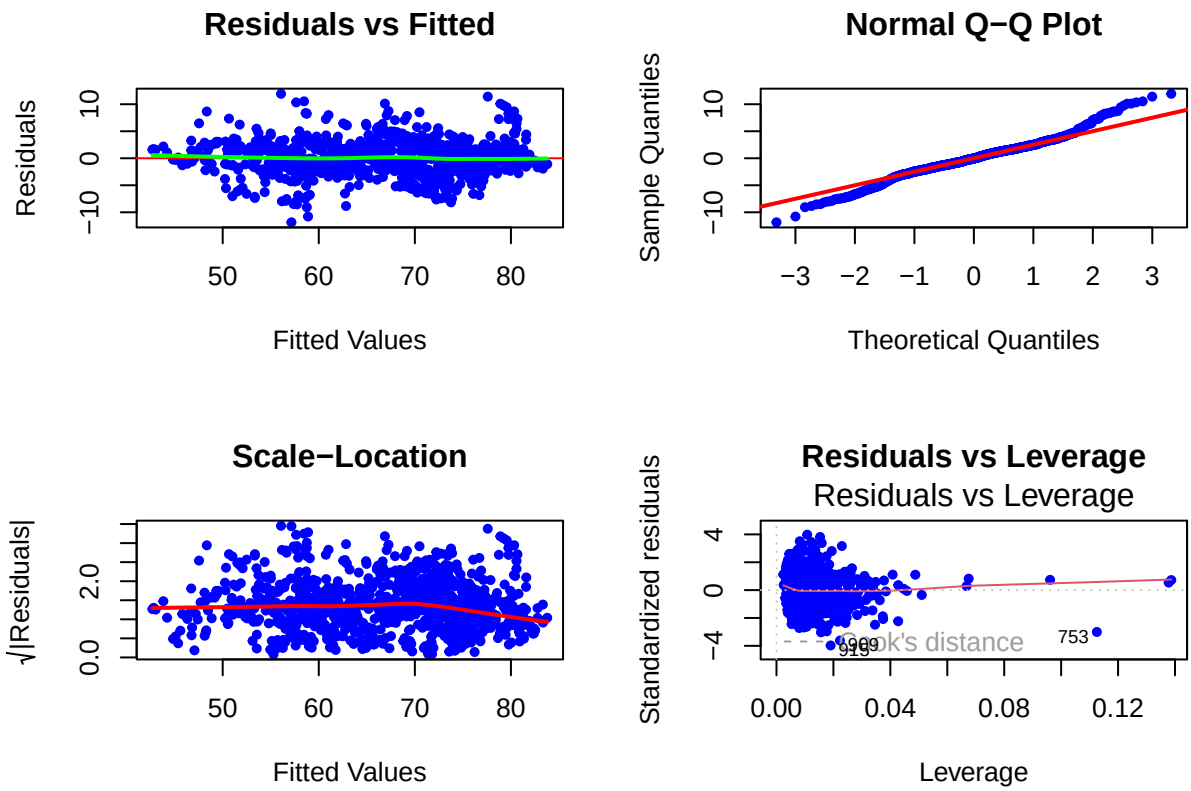
Model

Transformation and Main Effect Model

Before modeling, strongly skewed predictors—under-five deaths, percentage expenditure, population, and deaths per 1,000 live births due to HIV/AIDS (ages 0–4), motivate log-transformations: **(log(under.five.deaths + 1), log(percentage.expenditure + 1), log(population + 1), log(hiv.aids + 1)**. The main-effects model incorporated all cleaned and transformed predictors that has the following structure and summary:

```
Call:
lm(formula = life.expectancy ~ alcohol + measles + thinness.5.9.years +
    adult.mortality + log_under5 + total.expenditure + diphtheria +
    income.composition.of.resources + log_pctexp + log_pop +
    log_hiv + status, data = df_life_clean)
```

This model achieved an adjusted R^2 of about 0.89, but diagnostic plots showed residual curvature and mild heteroskedasticity, indicating that some nonlinear patterns and interactions were still missing.



After performing backward AIC, the reduced model's structure and summary look as follows.

```
Call:
lm(formula = life.expectancy ~ alcohol + thinness.5.9.years +
    adult.mortality + log_under5 + total.expenditure + diphtheria +
    income.composition.of.resources + log_pctexp + log_pop +
    log_hiv + status, data = df_life_clean)

Residuals:
     Min       1Q   Median       3Q      Max
-11.9088  -1.7093  -0.0774   1.7019  11.8858

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                     49.9203471  1.2078727  41.329  < 2e-16 ***
alcohol                         -0.1777715  0.0340436  -5.222 2.12e-07 ***
thinness.5.9.years              -0.0941892  0.0239172  -3.938 8.73e-05 ***
adult.mortality                 -0.0115088  0.0009252 -12.439  < 2e-16 ***
log_under5                      -0.2304824  0.0863320  -2.670  0.00770 **
total.expenditure                0.1243750  0.0428056   2.906  0.00374 **
diphtheria                       0.0134183  0.0046653   2.876  0.00410 **
income.composition.of.resources 31.7948090  1.2729475  24.977  < 2e-16 ***
log_pctexp                       0.2309936  0.0633769   3.645  0.00028 ***
log_pop                          0.1296581  0.0549217   2.361  0.01841 *
log_hiv                         -3.5430855  0.1514629 -23.392  < 2e-16 ***
statusDeveloping                -0.6299723  0.3611846  -1.744  0.08141 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.014 on 1095 degrees of freedom
Multiple R-squared:  0.8952,    Adjusted R-squared:  0.8941
F-statistic: 850.3 on 11 and 1095 DF,  p-value: < 2.2e-16
```

	Number of Predictors	R_squared	Adjusted R_squared	F-statistics	MSE
Main Effect Model	11	0.8952	0.8941	850.3	8.983529

Although backward AIC selection simplified the model and removed some predictors, these diagnostics motivated more complex modeling strategies.

Interaction Model

To capture potential synergistic effects between predictors, we introduced interaction terms informed by biological relationships and significance patterns from the main-effects model. The model has the following structure:

```
Call:
lm(formula = life.expectancy ~ alcohol + thinness.5.9.years +
    adult.mortality + log_under5 + total.expenditure + diphtheria +
    income.composition.of.resources + log_pctexp + log_pop +
    log_hiv + status + alcohol:thinness.5.9.years + adult.mortality:log_hiv +
    log_under5:income.composition.of.resources, data = df_life_clean)
```

After performing the same backward elimination based on AIC, the reduced model's structure and summary looked as follows:

```
Call:
lm(formula = life.expectancy ~ alcohol + thinness.5.9.years +
    adult.mortality + log_under5 + total.expenditure + diphtheria +
    income.composition.of.resources + log_pctexp + log_pop +
    log_hiv + alcohol:thinness.5.9.years + adult.mortality:log_hiv,
    data = df_life_clean)

Residuals:
     Min       1Q   Median       3Q      Max
-12.4209  -1.6613  -0.1003   1.6320  12.1972

Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                    52.2804956  1.2624002  41.414  < 2e-16 ***
alcohol                        -0.0506311  0.0415552  -1.218  0.22333
thinness.5.9.years             -0.0678307  0.0271803  -2.496  0.01272 *
adult.mortality                -0.0180094  0.0014081 -12.790  < 2e-16 ***
log_under5                     -0.2072826  0.0853891  -2.428  0.01536 *
total.expenditure               0.0998085  0.0421250   2.369  0.01799 *
diphtheria                      0.0122316  0.0045793   2.671  0.00767 **
income.composition.of.resources 28.9403972  1.3492626  21.449  < 2e-16 ***
log_pctexp                      0.2307123  0.0622284   3.708  0.00022 ***
log_pop                         0.1155746  0.0538611   2.146  0.03211 *
log_hiv                        -4.5905120  0.2507962 -18.304  < 2e-16 ***
alcohol:thinness.5.9.years     -0.0194678  0.0072723  -2.677  0.00754 **
adult.mortality:log_hiv         0.0037971  0.0006212   6.112 1.36e-09 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.954 on 1094 degrees of freedom
Multiple R-squared:  0.8994,    Adjusted R-squared:  0.8983
F-statistic: 814.8 on 12 and 1094 DF,  p-value: < 2.2e-16
```

	Number of Predictors	R_squared	Adjusted R_squared	F-statistics	MSE
Interaction Model	12	0.8994	0.8983	814.8	8.625933

Increasing values of R^2 and R^2_{adj} indicate adding interaction terms and then performing the backward elimination based on AIC was beneficial. However, the model remained overly complex relative to its predictive improvement.

Polynomial-Interaction Model

To address residual curvature without overcomplicating the model, we built a polynomial-interaction model by adding a quadratic term for adult mortality and keeping only the two meaningful interactions (alcohol $\times$ thinness and adult mortality $\times$ log(HIV)). This improved the nonlinear fit, and all retained predictors had VIF values below 4.

The final model is:

```
Call:
lm(formula = life.expectancy ~ alcohol + thinness.5.9.years +
    log_under5 + total.expenditure + diphtheria + income.composition.of.resources +
    log_pctexp + log_pop + log_hiv + adult.mortality + I(adult.mortality^2) +
    alcohol:thinness.5.9.years + adult.mortality:log_hiv, data = df_life_clean)

Residuals:
     Min       1Q   Median       3Q      Max
-12.3399  -1.5609  -0.1717   1.2457   9.8048

Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                    5.508e+01  1.117e+00  49.295  < 2e-16 ***
alcohol                        8.651e-02  3.720e-02   2.325  0.02024 *
thinness.5.9.years            -4.347e-02  2.386e-02  -1.822  0.06878 .
log_under5                    -2.268e-01  7.486e-02  -3.029  0.00251 **
total.expenditure              1.053e-01  3.693e-02   2.852  0.00442 **
diphtheria                     8.812e-03  4.019e-03   2.193  0.02853 *
income.composition.of.resources 2.473e+01  1.205e+00  20.514  < 2e-16 ***
log_pctexp                     1.218e-01  5.488e-02   2.220  0.02663 *
log_pop                        9.319e-02  4.723e-02   1.973  0.04876 *
log_hiv                       -6.574e+00  2.454e-01 -26.786  < 2e-16 ***
adult.mortality                7.761e-03  1.880e-03   4.129 3.91e-05 ***
I(adult.mortality^2)          -1.185e-04  6.519e-06 -18.181  < 2e-16 ***
alcohol:thinness.5.9.years    -3.662e-02  6.445e-03  -5.682 1.71e-08 ***
log_hiv:adult.mortality        2.024e-02  1.056e-03  19.172  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.59 on 1093 degrees of freedom
Multiple R-squared:  0.9227,    Adjusted R-squared:  0.9218
F-statistic:  1004 on 13 and 1093 DF,  p-value: < 2.2e-16
```

The expanded model underwent backward AIC selection with the following summary:

	Number of Predictors	R_squared	Adjusted R_squared	F-statistics	MSE
Polynomial Model	13	0.9227	0.9218	1004	6.622935

The R^2 value increases to 0.923 and the R^2_{adj} value increases to 0.922, while reducing the number of predictors.

Model Selection

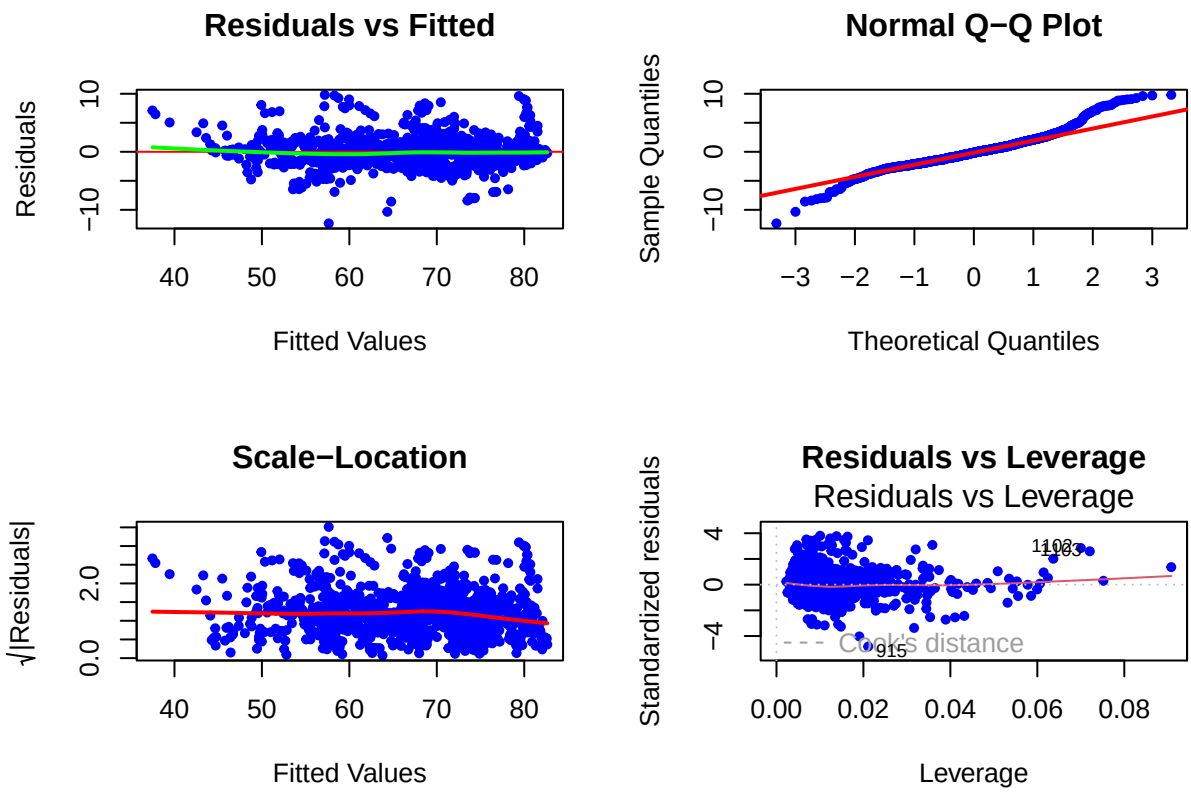
After building the models, we performed model selection based on the following factors: AIC, BIC, adjusted R^2, multiple R^2, and PRESS. Below is a comprehensive table that summarizes the findings:

	Main Effect Model	Reduced Interaction Model	Polynomial Model
R^2	0.8951983	0.89937	0.922737
R^2_{adj}	0.8941455	0.8982662	0.921818
C_p	24	26	28
AIC	5597.83	5554.864	5264.356
BIC	5662.952	5624.995	5339.497
$PRESS$	10184.54	9805.151	7550.819

The result is clear, the polynomial-interaction model has the highest multiple and adjusted R^2, the lowest AIC, BIC and the smallest PRESS. In terms of Mallow's C_p, the p' value for each model is 12, 13, and 14

respectively. While the main effect model has a C_p closest to its p' value, the polynomial-interaction model performs much better on all other criteria. Therefore, it is clear that the polynomial-interaction model is the best preforming model here and is selected to be our final model.

Model Validation and Diagnostic



Residuals appeared randomly scattered, the Q–Q plot indicated acceptable normality, and heteroskedasticity improved after applying log-transformations. Overall, the model satisfied the LINE assumptions.

	MSPR	MSE_F
Polynomial Model	6.820974	6.707767

The MSPR was very close to the in-sample MSE_F, indicating that the final model is not overfitted and generalizes well to new data.

Key Findings

All other variables held constant,

- Every 1% increase in "Deaths per 1 000 live births HIV/AIDS (0-4 years)" is associated with -0.06574 years decrease in life expectancy. This is very significant and could be because the data set mostly consisted of developing countries, where death by HIV/AIDS from 0-4 years old is more prevalent and frequent cause of death at early years.
- A country's income composition of resources has a significant impact on life expectancy of their people. The difference between the average life expectancy of countries of the lower quartile and the upper quartile was 19.89 years.

Limitations and Future Research

- We did not analyse patterns in missingness and just deleted all observations involving entries of NA. If missingness is correlated with an explanatory variable, then deleting these observations would cause bias in our model. This also leads to many lost observations. In the future, we recommend identifying the type of missing data, that is, whether it is MCAR, MAR, or MNAR. Depending on the missing data type, could choose to fill in missing data with mean/median, or other estimations. Furthermore, removed observations could be reintroduced back into the data set after the multicolinearity check because we do not care about NA entries in variables we will not be using.
- The model is only applicable to populations above 100k as we deleted all observations under 100k population. In the future, we recommend finding data for populations under 100k, and also finding other data sets to combine with our original data set. This improves our model as other explanatory variables could be added to our model.
- As the final model includes multiple interaction terms, it is hard to isolate and interpret the effect of a single explanatory variable. While this does not impact the model's predicting power, it could be hard to get bigger picture insights towards how each explanatory variable is impacting life expectancy. A simple model could be considered to draw bigger picture insights.

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Name	Student ID	Tasks/Responsibilities
Kit Mak	1010787819	Data Cleaning; EDA; VIF check for multicollinearity; Key findings, Limitations and Future Directions; Writing the report and presentation
Vika Permata Joe	1010309649	Data Cleaning; Main model building: Checking LINE assumption; Backward elimination; Add interaction terms; Add polynomial terms; Model selection; Final model validation and diagnostic; Writing the report and presentation